D7043E

Task 1

CRISP-DM for Customer Personality Analysis

By Group 3

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1. Introduction

This report documents Task 1 of the Advanced Data Mining project, following the CRISP-DM framework. The focus is on Business Understanding, Data Understanding, and Data Preparation, including feature engineering and exploratory analysis. Modeling and evaluation, the later phases of CRISP-DM, will be addressed in Task 2.

The dataset used is the *Customer Personality Analysis* dataset, available on Kaggle: https://www.kaggle.com/datasets/imakash3011/customer-personality-analysis/data

It contains demographic attributes, product spending patterns, purchase channels, and campaign responses for 2,216 retail customers. The aim for Task 1 is to prepare this dataset for two machine learning goals:

1. **Classification (Goal A):** Predict whether a customer will respond positively to a campaign.
2. **Clustering (Goal B):** Segment customers into personas based on demographics and behavior

2. Business Understanding

2.1 Context

Marketing costs increase when campaigns are sent to customers unlikely to respond. This dataset allows us to study customer attributes, spending patterns, and past responses to target campaigns at customers who will respond favorably to them. After successful completion of this data analysis and model training (Task 2), we will give a business the power to find and maintain customers based on the curated campaigns sent to them.

2.2 Business goals

- **Classification:** Rank customers by probability of campaign response. Marketing can focus on top scorers to improve conversion rates.
- **Clustering:** Group customers into interpretable personas (e.g., “mid-age premium spenders,” “budget family shoppers”). Personas allow campaigns to be tailored at segment level.

2.3 Success criteria

For Task 1, success means delivering a cleaned dataset with engineered features and exploratory analysis that reveals patterns. For Task 2, success will be measured by ROC-AUC for classification and cluster separation interpretability for clustering.

3. Data Understanding

3.1 Source and content

The cleaned dataset (`cleaned_marketing_campaign.csv`) contains 2,216 records and 27 original features. They span:

- **Demographics:** Education, Age, Marital status (simplified to Has_Partner), Children at home.
- **Economy:** Income, Spending amounts by category.
- **Customer relationship:** Tenure in days, Recency since last purchase.
- **Campaign history:** Acceptance of 5 campaigns, overall response.
- **Purchasing behavior:** Number of purchases by channel, monthly web visits.

3.2 Data issues

The raw data contains missing values in Income, inconsistent strings in categorical fields, and outliers in spending. These were resolved in cleaning before feature engineering:

- Income missing values were imputed.
- The birth year was converted into Age and capped at reasonable ranges.
- Categories like Education and Marital_Status were standardized.
- Duplicates were removed.

4. Data Preparation

4.1 Feature engineering

From the cleaned dataset, the following groups of features were engineered:

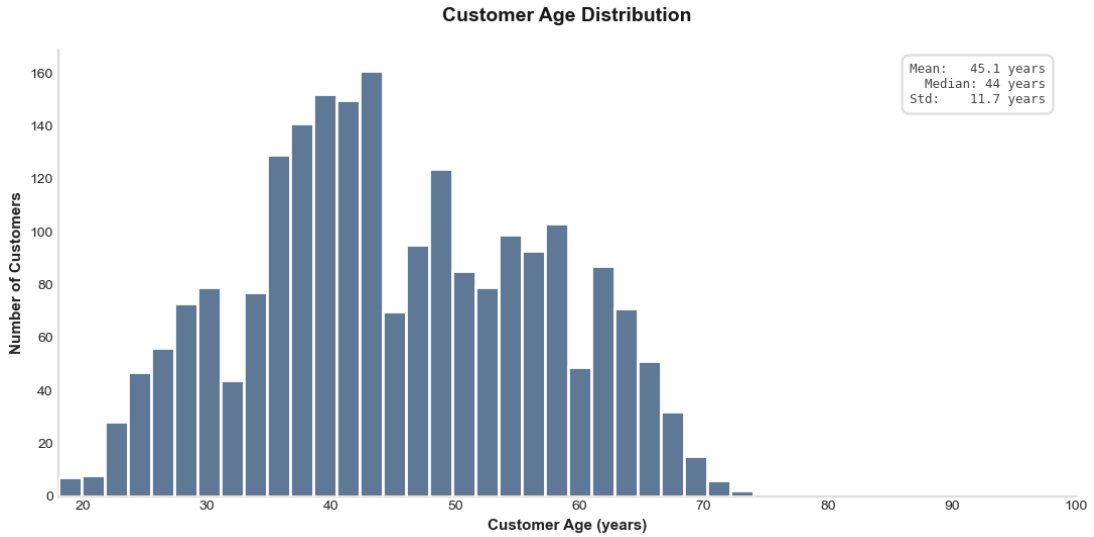
1. **Family & per-capita measures**
    - *Family_Size:* adults + children.
    - *Income_per_Capita, Spend_per_Capita, Dependency_Ratio.*
2. **Category mix & preference**
    - Shares of spending by product category (e.g., *Wines_Share*).
    - *Num_Categories_Active:* count of categories purchased.
    - *Category_Concentration:* degree of spending concentration.
    - *Premium_Tilt:* fraction of spend on Wines and Gold.
3. **Channel & engagement**
    - *Total_Purchases, Avg_Spend_per_Purchase.*
    - Shares of purchases by Web, Catalog, Store, and Deals.
4. **Campaign history**
    - *Past_Accepted, Ever_Accepted_Past, Campaign_Accept_Rate.*
5. **Recency & RFM scores**
    - *Tenure_Months, Frequency_pm.*
    - *Recency_Bin* into Hot/Warm/Cold.
    - *R_score, F_score, M_score, RFM_Total.*
6. **Transforms & interactions**
    - Log-transforms of Income and Spending.
    - Feature interactions: *Income_x_HasPartner, Age_x_WebShare.*
    - Binned variables: *Age_Bin, Income_q.*

The final processed dataset (`customers_featured.csv`) has 64 columns.

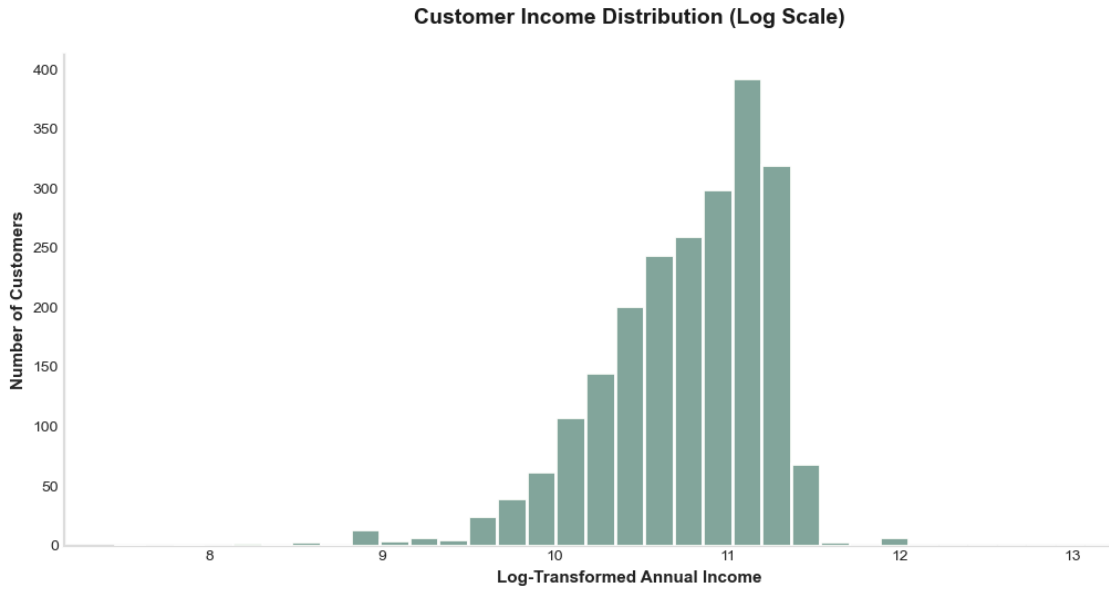
5. Exploratory Data Analysis

An exploratory analysis was conducted using the featured dataset, revealing some patterns present in the data. Ten key visuals were generated; four to five are included here, with others in the code notebook.

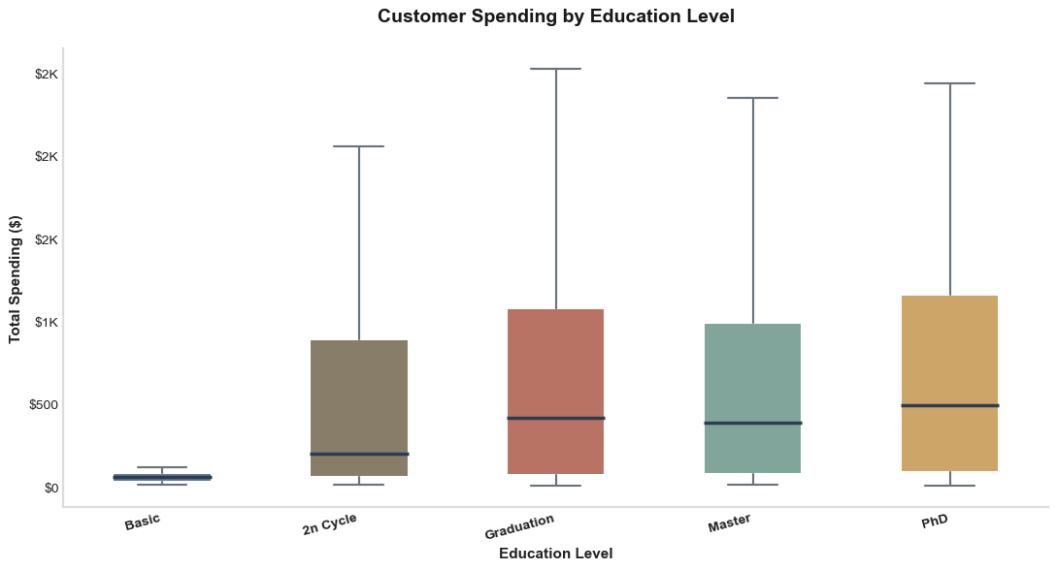
- The **age distribution** shows that most customers are between 30 and 60 years old and the median near 45 years old.



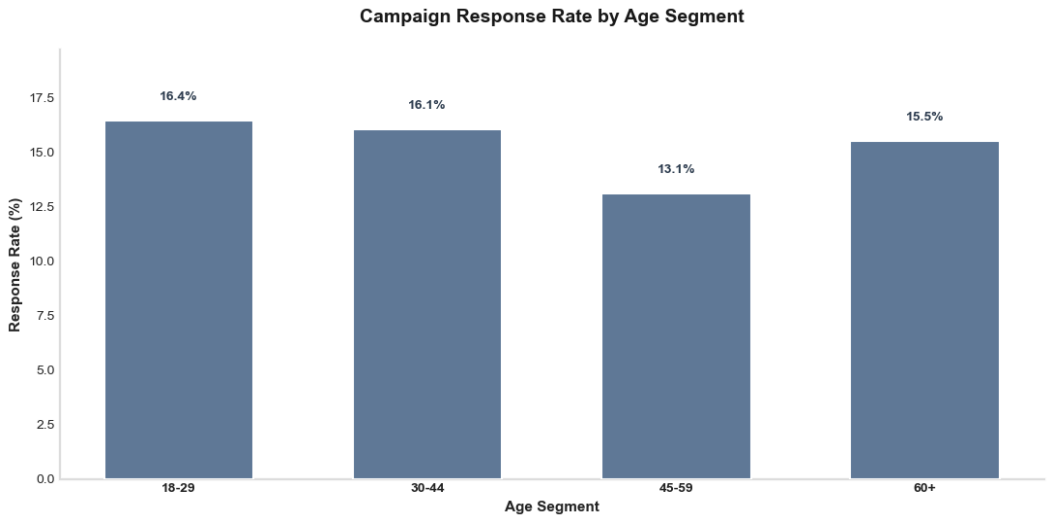
- **Income distribution** is highly skewed and using log transform normalizes it. As shown in the image below, the median income sits near mid-range values.



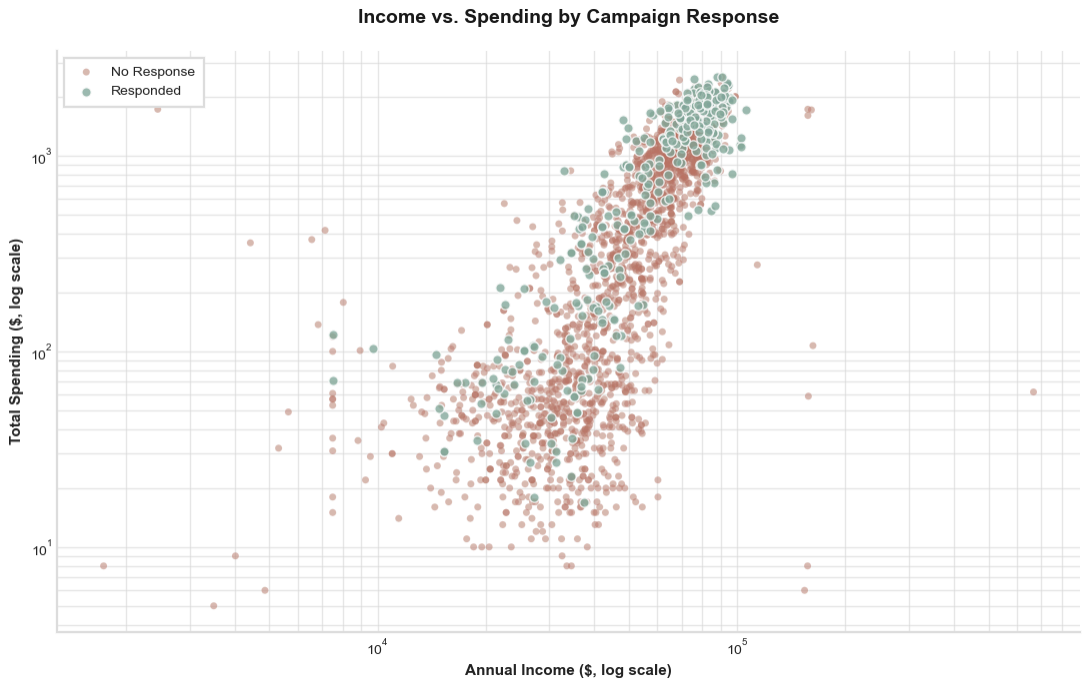
- The Customer's **Total Spending by Education** reveals that higher education groups show higher median total spend and wider spread than customers with lower education groups.



- **Response rate to Campaigns by Age Segment** shows higher rate of campaign responses among younger and mid-aged groups, with the best age segment being between 18-29 (16.4%).



- The explored **Income vs. Total_Spent, colored by Response** indicates that campaign responders are often mid-income with higher spending. It revealed from the dataset that there were a total of 333 responders and 1883 non-responders to campaigns



Other supporting visuals we created included: distribution of spending by category, RFM score distribution, and purchase channel shares.

6. Conclusion

In Task 1, we:

- Defined two AI goals: predicting campaign response (classification) and discovering personas (clustering).
- Cleaned the dataset and engineered features across family, product, channel, campaign, recency, and transformations.
- Produced exploratory plots that show meaningful structure relevant for both goals.

The processed dataset is now ready for Task 2, where we will train and compare classification models (e.g. a baseline mode trained with logistic + Random Forest compared to some Deep Neural Network) and clustering models (Autoencoder + K-Means, DEC).